

University of Illinois Department of Agricultural and Biological Engineering
Bioenvironmental and Structural Systems Lab
Final Report

Project Number: 24611
Test Date: November 25, 2024

Fan:
Make- Agrifan
Model- ZP-50-1.5ECS-100-1
Blade dia.- 53.2"
Orifice dia.- 53.4"

Motor:
Make- VSTEQ
Model- M56AA69A01
Hp- 1500 Watt
RPM- 730
Volts- 380
Amps- 2.85
Hz- 50
Phase- 3
S. F.- -
Housing:
Material- poly
Intake area- 53.5" dia.
Discharge- 53.4" dia.
Depth- 12"

Shutter: Butterfly Damper
Material- galvanized steel
Doors- 2
Columns- -
Door length- -
Location- exhaust
Guards:
Description- wire
Spacing- 2" x 2" / 2.8" concentric
Location- intake / exhaust
Discharge Cone:
Depth- 47.4"
Minor dia.- 53.4"
Major dia.- 69.3"

Blade:
Number- 6
Shape- propeller
Material- poly
Pitch- 32
Clearance- 0.1"

Drive Shaives:
Drive dia.- direct
Axle dia.- drive

Notes: *50 Hz test. 380VAC 3 phase input. 0 - 10 VDC speed control signal.
12" deep bell mouth intake shroud, 66" x 66" intake

Test Conditions:

T(wb) F: 56.9
T(db) F: 71.8 Barometric Pressure 29.03 (In. Hg)

							SI Units			
Static Pressure (in.H2O)	Airflow (cfm)	rpm	Volts	Amps	Watts	cfm/Watt	Static Pressure (Pa)	Airflow (m^3/hr.)	(m^3/hr)/W	W/1000m^3/hr
720 rpm (9.0 VDC)										
0.00	34600	720	380.2	2.33	1393	24.8	0	58700	42.2	24
0.05	33700	721	380.5	2.51	1503	22.4	12	57200	38.1	26
0.10	32700	721	380.4	2.68	1609	20.3	25	55600	34.6	29
0.15	31600	720	380.2	2.88	1738	18.2	37	53700	30.9	32
0.20	30500	720	380.7	3.07	1854	16.4	50	51800	27.9	36
0.25	29400	721	381.0	3.24	1970	14.9	62	49900	25.3	40
0.30	28200	721	381.3	3.42	2080	13.5	75	47900	23	43
0.35	27000	720	381.6	3.56	2170	12.5	87	45900	21.2	47
0.40	25700	720	381.9	3.70	2258	11.4	100	43700	19.4	52
0.45	24400	721	380.0	3.82	2325	10.5	112	41500	17.8	56
0.50	22500	721	380.4	3.91	2374	9.5	125	38200	16.1	62
685rpm (8.5 VDC)										
0.00	33100	686	378.6	1.99	1190	27.8	0	56300	47.3	21
0.05	32100	686	379.4	2.17	1305	24.6	12	54500	41.8	24
0.10	30900	686	379.9	2.37	1414	21.9	25	52600	37.2	27
0.15	29700	686	380.2	2.56	1533	19.4	37	50500	33	30
0.20	28600	686	380.0	2.73	1634	17.5	50	48600	29.8	34
0.25	27300	686	380.7	2.91	1748	15.6	62	46500	26.6	38
0.30	26200	686	380.8	3.04	1840	14.2	75	44500	24.2	41
0.35	24900	686	380.7	3.17	1918	13.0	87	42300	22.1	45
0.40	23400	686	381.3	3.29	1993	11.7	100	39800	20	50
0.45	21500	686	380.9	3.36	2042	10.5	112	36500	17.9	56
0.50	18400	686	381.3	3.35	2031	9.0	125	31200	15.4	65
600 rpm (7.5 VDC)										
0.00	28800	600	380.3	1.35	812	35.5	0	49000	60.4	17
0.05	27600	600	380.7	1.50	902	30.6	12	46900	52	19
0.10	26300	600	381.3	1.66	1005	26.2	25	44700	44.5	22
0.15	25000	600	381.8	1.82	1094	22.9	37	42500	38.8	26
0.20	23500	600	380.5	1.99	1199	19.6	50	40000	33.4	30
0.25	22100	600	380.7	2.11	1268	17.4	62	37600	29.6	34
0.30	20600	600	380.9	2.21	1323	15.6	75	35000	26.4	38
0.35	18700	600	381.1	2.27	1358	13.7	87	31700	23.3	43
0.40	14900	600	381.1	2.22	1335	11.1	100	25300	18.9	53
0.45	10000	600	380.6	2.05	1231	8.1	112	16900	13.7	73
0.50	8400	600	380.8	2.13	1284	6.5	125	14200	11.1	90
550 rpm (6.9 VDC)										
0.00	26400	551	378.1	1.10	653	40.4	0	44800	68.7	15
0.05	25200	551	379.6	1.23	741	34.0	12	42800	57.7	17
0.10	23800	552	380.0	1.38	828	28.8	25	40500	48.9	20
0.15	22200	551	380.4	1.51	916	24.2	37	37700	41.2	24
0.20	20600	551	380.9	1.63	987	20.9	50	35100	35.5	28
0.25	19000	551	381.0	1.72	1040	18.2	62	32200	31	32
0.30	16400	551	381.2	1.77	1075	15.3	75	27900	26	39
0.35	12300	551	380.9	1.68	1021	12.1	87	20900	20.5	49
0.40	8400	551	379.5	1.63	984	8.5	100	14300	14.5	69
0.45	6400	551	379.8	1.67	1008	6.3	112	10800	10.8	93
0.50	3700	551	379.9	1.69	1019	3.6	125	6300	6.2	161
500 rpm (6.3 VDC)										
0.00	24000	500	379.6	0.85	495	48.5	0	40800	82.5	12
0.05	22500	500	379.9	0.97	583	38.6	12	38300	65.6	15
0.10	20900	500	380.0	1.09	653	31.9	25	35400	54.3	18
0.15	19100	500	380.4	1.21	723	26.5	37	32500	45	22
0.20	17000	500	380.7	1.29	776	21.9	50	28900	37.3	27
0.25	14400	500	380.8	1.32	794	18.1	62	24400	30.8	32
0.30	9100	500	380.4	1.19	723	12.6	75	15400	21.3	47
0.35	6600	500	380.4	1.23	742	8.9	87	11300	15.2	66
0.40	3900	500	380.4	1.25	756	5.2	100	6700	8.8	113
0.45	1600	500	380.7	1.32	797	2.1	112	2800	3.5	285
0.51	0	500	381.2	1.46	885	0.0	126	0	0	-
450 rpm (5.7 VDC)										
0.00	21500	450	379.1	0.66	371	57.9	0	36500	98.4	10
0.05	19800	450	379.4	0.77	441	45.0	12	33700	76.5	13
0.10	17900	450	379.6	0.87	512	34.9	25	30300	59.2	17
0.15	15800	450	379.8	0.96	564	28.1	37	26900	47.7	21
0.20	13100	450	379.9	0.99	582	22.5	50	22200	38.2	26
0.25	7300	450	379.7	0.90	529	13.8	62	12400	23.4	43
0.30	5100	450	379.6	0.93	547	9.4	75	8700	16	63
0.35	2200	450	379.8	0.96	565	3.8	87	3700	6.5	154
0.41	0	450	380.0	1.07	646	0.0	101	0	0	-